

High-harmonic spin-current signatures of altermagnetic spin-group symmetry

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Spin point groups classify magnetic phases in the weak spin-orbit coupling regime and characterize the static properties of altermagnetic phases, but their dynamical consequences remain largely unexplored. Here, we derive selection rules for high-harmonic generation of charge and spin currents by extending dynamical symmetry to include spin point group operations. Since spin currents transform under both real and spin space operations, whereas charge currents transform only under real space operations, spin current selection rules can reveal magnetic information that is inaccessible to charge current harmonics. In a minimal altermagnetic model, an axis-aligned linearly polarized drive is non-diagnostic for distinguishing ferromagnetic and altermagnetic phases, although the antiferromagnetic phase is distinguished by the absence of the corresponding spin-current harmonics. A diagonal linearly polarized drive distinguishes the three SPG phases within the weak-SOC spin-group description, whereas a single-helicity circularly polarized drive provides a sharper spin-current-harmonic criterion for distinguishing them from magnetic-point-group mimics. These results establish spin current harmonics as a dynamical probe of spin group symmetry.

Introduction.— Symmetry is a key concept in physics, revealing universal principles and the diversity of emergent phenomena, especially in the classification of states of matter [1]. Recently, spin point group (SPG) symmetry, which combines crystallographic point group operations with independent spin space rotations, has been introduced to classify magnetic phases in the weak spin-orbit coupling (SOC) regime [2, 3]. This symmetry framework provides a natural way to distinguish altermagnets from conventional ferromagnets and antiferromagnets [4–6]. Altermagnets exhibit nonrelativistic momentum-dependent spin splitting despite their compensated collinear magnetic order, which is distinct from the relativistic spin splitting induced by SOC [7–9]. Owing to this electronic structure, altermagnets have attracted significant attention for spintronics applications [10–12], including spin filtering [13–17], spin-splitter torques [18–20], and spin current generation [21–26].

A promising route to probe the anisotropic spin splitting in altermagnets is high-harmonic generation (HHG) of charge and spin currents [27–34]. The properties of HHG, such as the allowed harmonic orders, are determined by dynamical symmetries, which combine symmetries of the undriven system with discrete time translations [34–43]. However, a selection-rule theory for charge and spin current HHG based on SPG symmetry remains undeveloped.

In this work, we extend dynamical symmetry by incorporating SPG operations and derive selection rules for charge and spin current HHG in altermagnets. We also demonstrate these selection rules by numerical HHG calculations in a minimal tight-binding model with D_{4h} symmetry. In this model, an axis-aligned linearly polarized drive gives identical selection rules for the ferromagnetic and altermagnetic phases and is therefore non-diagnostic. By contrast, a diagonal linearly polarized drive preserves a dynamical symmetry involving a spin space rotation and distinguishes the altermagnetic phase from ferromagnetic and antiferromagnetic phases. This linear-polarization diagnosis can, however, be ambiguous when compared with spin-orbit coupled systems described by magnetic point groups. A single-helicity circularly polarized drive resolves this ambiguity: the

spin space rotation in the SPG shifts the allowed spin current harmonics modulo four, whereas the corresponding magnetic point group operation flips the light helicity and is not a dynamical symmetry of the same drive.

Our results promote SPG symmetry from a classification principle for equilibrium magnetic structures to a predictive principle for driven nonlinear dynamics. In particular, we show that spin current HHG can access spin space symmetry information that is inaccessible to charge current HHG alone. This provides symmetry-based criteria for distinguishing altermagnetic phases from conventional ferromagnetic and antiferromagnetic phases, and more broadly establishes spin current HHG as a symmetry-resolved nonlinear probe of weak-SOC magnets. In addition, our framework clarifies how the selection rules originating from nonrelativistic altermagnetic spin splitting differ from those associated with relativistic SOC-induced spin splitting.

Dynamical symmetries for charge and spin currents.— A dynamical symmetry combines a spatial symmetry with a time translation. We write it as $(g; \tau)$, where g is a static symmetry operation and τ is a time translation. For the observable $O(t)$, the dynamical symmetry $G_\tau = (g; \tau)$ acts as

$$G_\tau O(t) G_\tau^{-1} = D(g) O(t + \tau), \quad (1)$$

where $D(g)$ is the representation of the spatial symmetry operation g for the observable O . Under this dynamical symmetry, the Fourier component $O^{(N)} = \int_0^T O(t) e^{-iN\Omega t} dt / T$, where T is the period of the driving field, satisfies

$$(I - e^{iN\Omega\tau} D(g)) O^{(N)} = 0, \quad (2)$$

where I is the representation of the identity operator. This relation gives the selection rule for the Fourier component of the observable $O^{(N)}$.

In the weak-SOC regime, collinear magnets are classified by SPGs [2, 4, 5]. In this paper, we use the notation of the SPG operation as $[g_s \parallel g_r]$, where g_s is the spin transformation and g_r is the space transformation. Type-I and type-II phases correspond to conventional ferromagnets and antiferromagnets,

whereas type-III phases, i.e., altermagnets, contain operations of the form $[C_2 \parallel g_r]$ that flip the collinear spin direction together with a nontrivial real space operation.

We focus on unitary SPG operations, because they directly encode nonrelativistic spin space symmetries in weak-SOC magnets. Antiunitary magnetic point group operations, relevant to relativistic SOC-induced spin textures, are discussed in the Supplemental Material (SM). In particular, under single-helicity CPL, $C_n\mathcal{T}$ does not by itself act as the same dynamical symmetry as $[C_2 \parallel C_n]$, because $\mathcal{T}$ flips the helicity of light.

The key ingredient for deriving HHG selection rules is the representation of SPG operations for charge and spin currents. The charge current transforms as $D_{J_c}([g_s \parallel g_r]) = D_v(g_r)$. The spin current transforms as $D_{J_s}([g_s \parallel g_r]) = D_v(g_r) \otimes D_s(g_s)$, where D_v and D_s are the vector representations of the space and spin transformations, respectively. Thus, the charge current is a scalar under spin transformations, whereas the spin current is a vector in spin space. Charge current selection rules are therefore determined only by space transformations, while spin current selection rules depend on both space and spin transformations. Consequently, spin current selection rules can differ between type-I/type-II and type-III magnets, whereas charge current selection rules are insensitive to whether the real space operation is accompanied by E or by C_2 in spin space.

Minimal model and methods.— As a minimal example, we consider a crystal with point group D_{4h} . The type-I magnet is given by $[E \parallel D_{4h}]$, while the type-II magnet is given by $[E \parallel D_{4h}] + [C_{2x} \parallel D_{4h}]$. For the same crystal, the type-III magnet is given by the SPG $[E \parallel D_{2h}] + [C_{2x} \parallel (D_{4h} - D_{2h})]$, which corresponds to a d -wave altermagnet. The D_{4h} model provides a minimal symmetry setting for d -wave altermagnetism and serves as a representative framework for rutile-type altermagnetic candidates, whose material realization remains under active debate, as exemplified by RuO_2 [4, 44, 45]. This setting allows us to isolate the symmetry consequences of d -wave altermagnetic order in nonlinear charge and spin current responses.

As a minimal model for a d -wave altermagnet, we use the following Hamiltonians on a two-sublattice square lattice:

$$H_I(\vec{k}) = \epsilon_0(\vec{k})I_2 \otimes I_2 + d_z(\vec{k})I_2 \otimes \tau_z + \Delta\sigma_z \otimes I_2, \quad (3)$$

$$H_{II}(\vec{k}) = \epsilon_0(\vec{k})I_2 \otimes I_2 + M\sigma_z \otimes \tau_z, \quad (4)$$

$$H_{III}(\vec{k}) = \epsilon_0(\vec{k})I_2 \otimes I_2 + d_z(\vec{k})I_2 \otimes \tau_z + M\sigma_z \otimes \tau_z, \quad (5)$$

where $\sigma_{x,y,z}$ and $\tau_{x,y,z}$ are the Pauli matrices for the spin and sublattice degrees of freedom, respectively, and $\epsilon_0(\vec{k}) = t(2 - \cos(ak_x) - \cos(ak_y))$ and $d_z(\vec{k}) = t'(\cos(ak_x) - \cos(ak_y))$. In the type-II case, the operation $[C_{2x} \parallel E]$ should be understood as including the sublattice-exchanging translation that flips τ_z , so that the staggered exchange term $M\sigma_z \otimes \tau_z$ remains invariant.

To verify these selection rules for charge and spin current HHG, we use the phenomenological equation of motion (EoM)

for the density matrix $\rho_{\vec{k}}(t)$ given by

$$\frac{d}{dt}\rho_{\vec{k}}(t) = -i[H(\vec{k} + \vec{A}(t)), \rho_{\vec{k}}(t)] - \frac{1}{T_2}(\rho_{\vec{k}}(t) - \rho_{\vec{k}}^{\text{eq}}), \quad (6)$$

where T_2 is the relaxation time and $\rho_{\vec{k}}^{\text{eq}}$ is the equilibrium density matrix. The relaxation term is assumed to preserve the dynamical symmetries, so that it does not modify the symmetry-derived selection rules. The Hamiltonian $H(\vec{k})$ is given by $H_I(\vec{k})$, $H_{II}(\vec{k})$, and $H_{III}(\vec{k})$ for the type-I, type-II, and type-III magnets, respectively. The driving external field $\vec{A}(t)$ is given by $\vec{A}(t) = \vec{A}_0(t)f_{\text{env}}(t)$, where $f_{\text{env}}(t)$ is the envelope function and $\vec{A}_0(t) = A_0\vec{e}_{\parallel}\cos(\Omega t)$ for the linearly polarized light (LPL) and $\vec{A}_0(t) = A_0(\cos(\Omega t), \sin(\Omega t))/\sqrt{2}$ for the single-helicity circularly polarized light (CPL). The unit vector $\vec{e}_{\parallel}$ denotes the polarization direction for the LPL, and the amplitude A_0 is given by $A_0 = E_0/\Omega$, where E_0 is the electric field amplitude and Ω is the frequency. The charge and spin current expectation values, $\langle J_{e,v} \rangle(t)$ and $\langle J_{s,\mu,v} \rangle(t)$, are computed from the density matrix using the current operators $J_{e,v} = -\partial_{k_v}H$ and $J_{s,\mu,v} = \{\sigma_{\mu}, J_{e,v}\}$, and their N -th harmonic components $J_{e,v}^{(N)}$ and $J_{s,\mu,v}^{(N)}$ are obtained by Fourier transformation. The conventional factor 1/2 in the spin-current operator is omitted, since it only rescales the spectra and does not affect the selection rules. Since the z -spin component is conserved in all three magnetic phases considered here, we focus below on the z -polarized spin current $J_{s,z,v}$.

Results and Discussion.— We first summarize the selection rules derived from SPG dynamical symmetries, as shown in Table I. This table shows the allowed harmonic orders of charge and spin currents for the type-I, type-II, and type-III magnets under LPL_x, LPL_[110], and single-helicity CPL. The z -polarized spin current in type-II magnets is forbidden for all orders. This is a direct consequence of the fact that the type-II SPG contains both $[E \parallel g_r]$ and $[C_{2x} \parallel g_r]$ operations for all g_r , which impose incompatible constraints on the spin current. For LPL polarized along the x -direction (LPL_x), the dynamical symmetry $([E \parallel C_{2y}]; T/2)$ and $([E \parallel C_{2x}]; 0)$ are common to the type-I, type-II, and type-III magnets. The symmetry $([E \parallel C_{2y}]; T/2)$ imposes the selection rule for longitudinal charge and spin currents, $J_{e,\parallel}^{(2m+1)}$ and $J_{s,z,\parallel}^{(2m+1)}$, and the symmetry $([E \parallel C_{2x}]; 0)$ imposes the prohibition of the transverse components $J_{e,\perp}^{(N)}$ and $J_{s,z,\perp}^{(N)}$ for all orders. Therefore, both charge and spin current harmonics obey the same selection rules for type-I and type-III magnets, and LPL_x does not distinguish their SPG structures.

The selection rules for LPL along the $[110]$ direction (LPL_[110]) are derived by the dynamical symmetries $([E \parallel C_{2,\bar{1}10}]; T/2)$, $([E \parallel C_{2,110}]; 0)$ for type-I magnet and $([C_{2x} \parallel C_{2,\bar{1}10}]; T/2)$, $([C_{2x} \parallel C_{2,110}]; 0)$ for type-III magnet. These operations impose the same constraint on the charge current, because the charge current is insensitive to spin space operations. In particular, the odd-order z -polarized spin-current harmonic appears in the longitudinal channel for the type-I magnet, whereas it appears in the transverse channel for the type-III magnet.

TABLE I. Summary of selection rules for the charge current and the z -polarized spin current. Only symmetry-allowed nonzero components within this set are listed; components not shown are forbidden. Each entry lists charge-current and spin-current harmonics separated by a comma. A dash denotes z -polarized spin-current harmonics forbidden for all orders. Here $m \in \mathbb{Z}$, and $J_{\pm} = J_x \pm iJ_y$ for CPL.

Drive	Type-I	Type-II	Type-III
LPL _x	$J_{e,\parallel}^{(2m+1)}, J_{s,z,\parallel}^{(2m+1)}$	$J_{e,\parallel}^{(2m+1)}, -$	$J_{e,\parallel}^{(2m+1)}, J_{s,z,\parallel}^{(2m+1)}$
LPL _[110]	$J_{e,\parallel}^{(2m+1)}, J_{s,z,\parallel}^{(2m+1)}$	$J_{e,\parallel}^{(2m+1)}, -$	$J_{e,\parallel}^{(2m+1)}, J_{s,z,\perp}^{(2m+1)}$
CPL	$J_{e,\pm}^{(4m\pm 1)}, J_{s,z,\pm}^{(4m\pm 1)}$	$J_{e,\pm}^{(4m\pm 1)}, -$	$J_{e,\pm}^{(4m\pm 1)}, J_{s,z,\pm}^{(4m\pm 1)}$

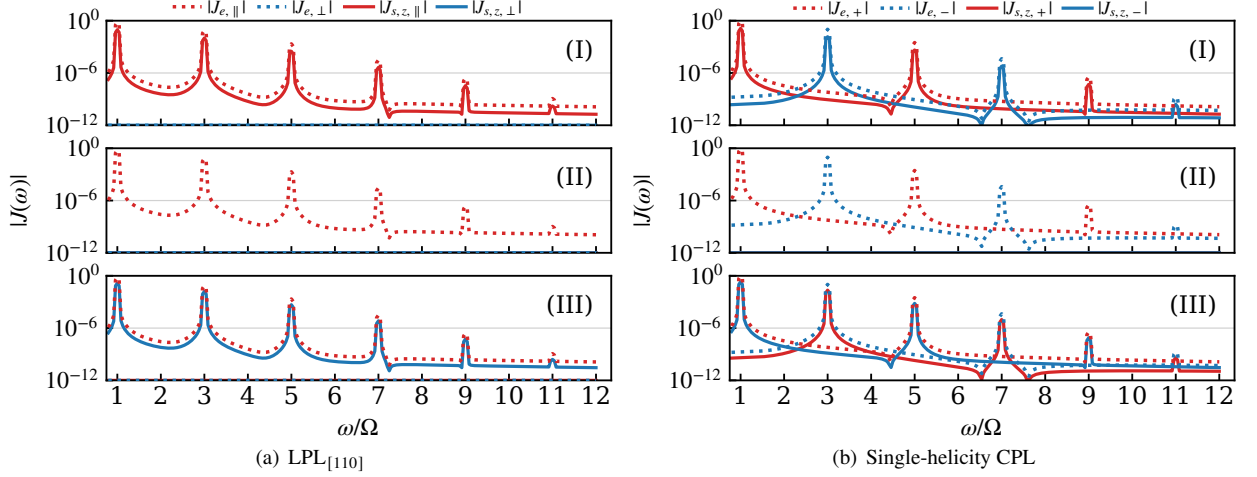


FIG. 1. HHG of charge and spin currents in the type-I (top panel), type-II (middle panel), and type-III (bottom panel) magnets under LPL_[110] (a) and CPL (b). The vertical axis shows the absolute value of the Fourier components of charge and spin currents. The horizontal axis shows the frequency ω normalized by the driving frequency Ω . The dotted lines correspond to the charge current, and the solid lines correspond to the spin current. (a) The polarization direction of the LPL_[110] is set as $\vec{e}_{\parallel} = (1, 1)/\sqrt{2}$. The red and blue lines correspond to the parallel and perpendicular components of the charge (dotted) and spin (solid) currents, respectively. (b) The red and blue lines correspond to the right-handed (+) and left-handed (-) circular components of the charge (dotted) and spin (solid) currents, respectively.

The LPL selection rule, however, should not be overinterpreted as a unique fingerprint of the SPG. A magnetic point group can produce an LPL constraint similar to that generated by the spin point group operation [$C_2 \parallel C_2$]. Consequently, LPL alone cannot unambiguously distinguish a type-III SPG response from a spin-orbit coupled system whose magnetic point group produces a similar LPL selection rule. As shown in the SM [46], the LPL selection rules from magnetic point groups depend on the relative orientation between the crystal axes and the linear polarization. Thus, LPL alone is not an unambiguous probe of the SPG origin.

The single-helicity CPL resolves this ambiguity more directly. For a single-helicity CPL, the fourfold dynamical symmetry gives modulo-four selection rules. The type-I magnet has $([E \parallel C_{4z}]; -T/4)$, whereas the type-III magnet has $([C_{2x} \parallel C_{4z}]; -T/4)$. The two operations again give the same constraint on the charge current but different constraints on the spin current, resulting in a modulo-four shift of the allowed spin current harmonics between the type-I and type-III magnets. For the type-II magnet, the simultaneous presence of both constraints prohibits the corresponding spin current harmonics.

Within the fixed-geometry, fixed-helicity CPL configuration considered here, this modulo-four spin-current selection rule

is not reproduced by the corresponding magnetic point group operation. In the present geometry, the relevant magnetic point group candidates are an ordinary fourfold rotation and a magnetic fourfold rotation $C_{4z}\mathcal{T}$, where $\mathcal{T}$ denotes time reversal. The ordinary fourfold rotation gives the type-I selection rule, because spin is locked to real space and no independent spin space operation is available. The magnetic fourfold rotation, on the other hand, involves time reversal and flips the helicity of CPL; in the fixed incident geometry considered here, this helicity flip is not compensated by any additional operation and therefore is not a dynamical symmetry of the same drive. Therefore, the type-III CPL selection rule reflects the independent spin space operation of the SPG and provides a sharper fingerprint of a type-III magnet than LPL does.

We finally verify these symmetry-derived predictions using real-time density-matrix simulations of a minimal weak-SOC tight-binding model. Figure 1(a) and (b) show the numerical results for LPL_[110] and CPL, respectively. The results for LPL_x are shown in the SM [46]. The numerical HHG spectra reproduce the predicted selection rules: LPL_[110] and CPL reveal the type-III SPG symmetry through spin current harmonics. In particular, the CPL spectra show the predicted modulo-four shift of the allowed spin current harmonics, while

the charge current harmonics remain insensitive to the distinction between the type-I and type-III magnets. These results establish spin current HHG as a symmetry-resolved nonlinear probe of weak-SOC magnetic phases.

Experimentally, the spin-current harmonics considered here would most naturally be accessed indirectly through spin-to-charge conversion in a heterostructure geometry. For example, the altermagnet may be interfaced with a strong-SOC metal such as Pt, where the injected ac spin current is converted into an ac charge current by the inverse spin Hall effect [47–49], or with an inversion-asymmetric interface [50]. Such conversion schemes are standard in spin-pumping experiments and have also been proposed for detecting high-harmonic spin pumping in altermagnets [34]. In this work, we focus on the intrinsic symmetry selection rules of the spin current before interfacial conversion; whether a given detector preserves or masks these selection rules depends on the spin-to-charge conversion tensor and on the interface symmetry. At leading order, a linear and time-independent conversion, $J_{c,i}^{(N)} = \sum_{\mu,\nu} \Lambda_{i\mu\nu} J_{s,\mu,\nu}^{(N)}$, preserves the harmonic order, although the measured component and helicity depend on the conversion tensor and interface symmetry. A quantitative detector-level prediction requires an explicit treatment of the conversion tensor and interface symmetry and is left for future work.

Conclusion.— We have extended dynamical-symmetry selection rules for high-harmonic generation to SPGs and applied them to charge and spin current responses in weak-SOC magnets. Because charge currents transform only under real space operations, whereas spin currents transform under both real space and spin space operations, spin current HHG can detect symmetry information that is invisible in charge current HHG. For the real space operation $g_r \in \mathbf{G} - \mathbf{H}$, where $\mathbf{G}$ and $\mathbf{H}$ are the crystallographic point group and its subgroup, respectively, the SPG operation $[E \parallel g_r]$ and $[C_{2x} \parallel g_r]$ give the same selection rule for the charge current, while they give conflicting selection rules for the spin current. Thus, the selection rules for spin current harmonics are interchanged or shifted between type-I and type-III magnets, depending on the drive symmetry, while the corresponding harmonics are prohibited for all orders in the type-II magnet. In particular, a single-helicity CPL drive gives a modulo-four shift of the allowed spin current harmonics between type-I and type-III magnets, while the corresponding magnetic point group operation does not give the same selection rule for the fixed-helicity CPL geometry considered here. Therefore, spin current HHG can distinguish weak-SOC magnetic phases. We have confirmed these symmetry predictions using density-matrix simulations of a minimal d -wave altermagnetic tight-binding model. These results promote SPG symmetry from a classification principle for equilibrium magnetic structures to a predictive principle for driven nonlinear dynamics, and establish spin current HHG as a symmetry-resolved nonlinear probe of altermagnets.

Data availability.— The data and code supporting the findings of this study are not currently publicly available. They will be deposited in the NAGOYA Repository, the institutional repository of Nagoya University, following publication. They

are available from the corresponding author upon reasonable request.

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SUPPLEMENTAL MATERIAL

Antiunitary dynamical symmetries

For the antiunitary dynamical symmetry $A_\tau = (g\mathcal{T}; \tau)$, a representative convention-dependent form of the constraint is

$$(I - \eta_O e^{iN\Omega\tau} D(g)K) O^{(N)} = 0, \quad (7)$$

where $\mathcal{T}$ is the time-reversal operation, g is the unitary point group operation, $O^{(N)}$ is the N -th high-harmonic component of the observable O , η_O is the parity of the physical quantity O under the time-reversal operation, and K is the complex conjugation operator. The precise phase form depends on the time origin and Fourier convention.

In this work, we focus on dynamical symmetries generated by unitary operations. Antiunitary dynamical symmetries can also impose selection rules; however, their explicit form depends on the choice of the time origin because of the complex conjugation involved in antiunitary operations. For this reason, and because our main objective is to isolate the consequences of nonrelativistic spin-group symmetries, we formulate the main analysis in terms of unitary spin point-group operations.

This formulation is sufficient to characterize symmetry constraints imposed by magnetic order in the weak-SOC regime, where spin-space and real-space operations can be treated independently. In particular, a type-III altermagnet is characterized by a spin point-group operation of the form $[C_2 \parallel C_n]$, where C_2 flips the collinear spin direction and C_n is an n -fold real-space rotation. As shown below, this operation gives rise to characteristic selection rules for charge- and spin-current HHG.

Nevertheless, antiunitary magnetic point group symmetries are important when one compares altermagnetic spin splitting with relativistic spin-orbit-coupled spin splitting. A useful relativistic analogue of a $[C_2 \parallel C_n]$ altermagnet is a centrosymmetric SOC system whose spin texture is constrained by an antiunitary magnetic point-group operation such as $C_n\mathcal{T}$. The presence of inversion symmetry is important here because it allows the relativistic spin splitting to be even in momentum space, making it comparable to the d -wave- or g -wave-like spin splitting of altermagnets. In such a system, the antiunitary operation $C_n\mathcal{T}$ can relate opposite spin textures at momenta connected by C_n , in analogy with the spin point-group operation $[C_2 \parallel C_n]$ in the weak-SOC altermagnet. However, under a fixed single-helicity circularly polarized drive, time reversal $\mathcal{T}$ flips the helicity of the light. Since a proper rotation C_n does not flip the helicity, the combined operation $C_n\mathcal{T}$ does not generally define a dynamical symmetry of the single-helicity CPL-driven system, unless it is accompanied by an additional helicity-reversing operation or by changing the incident helicity.

Therefore, under single-helicity circularly polarized light, the characteristic selection rules discussed here originate from the unitary spin point-group operation $[C_2 \parallel C_n]$, rather than from the antiunitary magnetic point-group operation $C_n\mathcal{T}$. This distinction provides a symmetry-based criterion for separating nonrelativistic altermagnetic spin splitting from its relativistic SOC-induced analogue.

Minimal model

Here, we introduce a minimal model for the type-I, type-II, and type-III magnets. The operations in the spin point group are denoted as $[g_s \parallel g_r]$, where g_s is the spin transformation and g_r is the space transformation. The type-I magnet corresponding to the ferromagnetic phase is described by $[E \parallel \mathbf{G}]$, the type-II magnet corresponding to the antiferromagnetic phase is described by $[E \parallel \mathbf{G}] + [C_2 \parallel \mathbf{G}]$, and the type-III magnet is described by $[E \parallel \mathbf{H}] + [C_2 \parallel (\mathbf{G} - \mathbf{H})]$, where $\mathbf{G}$ is the group of lattice symmetries and $\mathbf{H}$ is a subgroup of $\mathbf{G}$. For the type-II magnet in particular, the symmetry operation with spin flip is combined with the space translation that connects the two sublattices. A type-III magnet is an altermagnet, and the altermagnetic phase is characterized by a spin point group that combines the spin transformation C_2 with the space transformations in the coset $\mathbf{G} - \mathbf{H}$.

As the simplest example, we consider a crystal with point group D_{4h} , and the type-I magnet is given by $[E \parallel D_{4h}]$, the type-II magnet is given by $[E \parallel D_{4h}] + [C_{2x} \parallel D_{4h}]$, and the type-III magnet is given by $[E \parallel D_{2h}] + [C_{2x} \parallel (D_{4h} - D_{2h})]$. This type-III SPG describes a d -wave altermagnet, and the spin splitting has a $d_{x^2-y^2}$ -wave form in momentum space.

To describe ferromagnetic, antiferromagnetic, and altermagnetic phases within a common framework, we consider a spinful two-sublattice system with sublattice degrees of freedom A and B . The Hamiltonian is represented by a 4×4 matrix in the basis of $(|A, \uparrow\rangle, |B, \uparrow\rangle, |A, \downarrow\rangle, |B, \downarrow\rangle)^T$. The irreducible decomposition of the Hamiltonian is given by $\sigma_\mu \otimes \tau_\nu$, where σ_μ and τ_ν are the Pauli matrices for the spin and sublattice degrees of freedom, respectively. For the sublattice space, the representation of the symmetry operation g is given by $D_\tau(g) = \tau_0$ for $g \in D_{2h}$ and $D_\tau(g) = \tau_x$ for $g \in (D_{4h} - D_{2h})$. Under these constraints, the

Hamiltonian for the type-I, type-II, and type-III magnets is given by

$$H_I(\vec{k}) = \epsilon_0(\vec{k})I_2 \otimes I_2 + d_z(\vec{k})I_2 \otimes \tau_z + \Delta\sigma_z \otimes I_2, \quad (8)$$

$$H_{II}(\vec{k}) = \epsilon_0(\vec{k})I_2 \otimes I_2 + M\sigma_z \otimes \tau_z, \quad (9)$$

$$H_{III}(\vec{k}) = \epsilon_0(\vec{k})I_2 \otimes I_2 + d_z(\vec{k})I_2 \otimes \tau_z + M\sigma_z \otimes \tau_z, \quad (10)$$

where $\epsilon_0(\vec{k}) = t(2 - \cos(ak_x) - \cos(ak_y))$ and $d_z(\vec{k}) = t'(\cos(ak_x) - \cos(ak_y))$. These models correspond to tight-binding models on a staggered square lattice with only intra-sublattice hoppings t and t' , where t and t' are sublattice-independent and sublattice-dependent hopping parameters, respectively. The ferromagnetic order parameter Δ and the antiferromagnetic order parameter M represent uniform and staggered exchange fields, respectively.

In Fig. 2, we show the band structure of the model Hamiltonian for the type-I, type-II, and type-III magnets. The type-I magnet has uniform spin splitting throughout the Brillouin zone due to the ferromagnetic order denoted by Δ . The type-II magnet is the antiferromagnetic phase without spin splitting throughout the Brillouin zone. The type-III magnet has the d -wave spin splitting due to the altermagnetic order.

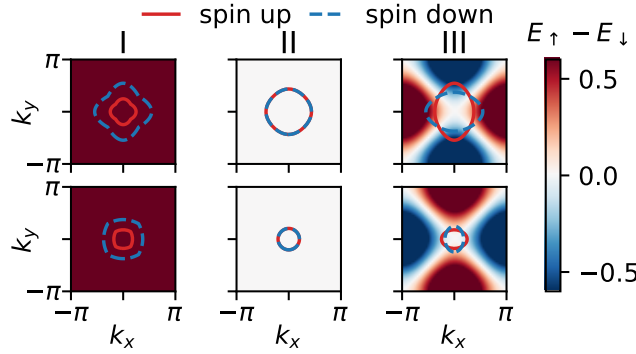


FIG. 2. Band structure of the model Hamiltonian for the type-I, type-II, and type-III magnets denoted by the top label. The upper and lower panels correspond to the lower and upper two bands, respectively. The spin up and down components are denoted by the red solid and blue dashed lines, respectively. The color map shows the spin splitting $E_\uparrow - E_\downarrow$. In our calculation, we set lattice constants as $a_x = a_y = 1.0$, hopping parameters as $t = 1.0$, $t' = 0.3$, and chemical potential as $\mu = 0.5$. For the type-I magnet, we set the ferromagnetic order parameter as $\Delta = 0.3$. For the type-II and type-III magnets, we set the antiferromagnetic order parameter as $M = 0.3$.

Time evolution of the density matrix

In the main text, we use the phenomenological equation of motion (EoM) for the density matrix $\rho_{\vec{k}}(t)$ given by

$$\frac{d}{dt}\rho_{\vec{k}}(t) = -i[H(\vec{k} + \vec{A}(t)), \rho_{\vec{k}}(t)] - \frac{1}{T_2}(\rho_{\vec{k}}(t) - \rho_{\vec{k}}^{\text{eq}}), \quad (11)$$

where T_2 is the relaxation time and $\rho_{\vec{k}}^{\text{eq}}$ is the equilibrium density matrix. The relaxation term in Eq. (11) can be motivated by the following simple Lindblad form. The Lindblad equation is

$$\frac{d}{dt}\rho_{\vec{k}}(t) = -i[H(\vec{k} + \vec{A}(t)), \rho_{\vec{k}}(t)] + \sum_i \left[L_i \rho_{\vec{k}}(t) L_i^\dagger - \frac{1}{2} \{ L_i^\dagger L_i, \rho_{\vec{k}}(t) \} \right]. \quad (12)$$

For the dissipative part, we consider the diagonal basis of the equilibrium density matrix $\rho_{\vec{k}}^{\text{eq}} = \sum_n p_n |n\rangle \langle n|$, where $\sum_n p_n = 1$. In this basis, the Lindblad operators are assumed to take the form of $L_i := L_{mn} = \sqrt{\gamma p_m} |m\rangle \langle n|$. Therefore, using trace preservation, one obtains

$$\frac{d}{dt}\rho_{\vec{k}}(t) = -i[H(\vec{k} + \vec{A}(t)), \rho_{\vec{k}}(t)] - \gamma (\rho_{\vec{k}}(t) - \rho_{\vec{k}}^{\text{eq}}). \quad (13)$$

Thus, one obtains the EoM (11) with $T_2 = 1/\gamma$. This Lindblad construction is used only to justify the relaxation-to-equilibrium structure phenomenologically; the selection rules derived in this work do not rely on the microscopic form of the relaxation channel, provided that the dissipative dynamics respects the same dynamical symmetries.

The Hamiltonian $H(\vec{k})$ is given by $H_I(\vec{k})$, $H_{II}(\vec{k})$, and $H_{III}(\vec{k})$ for the type-I, type-II, and type-III magnets, respectively. The driving external field $\vec{A}(t)$ is given by $\vec{A}(t) = \vec{A}_0(t)f_{\text{env}}(t)$. The envelope function is

$$f_{\text{env}}(t) = \begin{cases} 1, & |t| \leq 120, \\ \frac{1}{2} \left[1 + \cos \left(\pi \frac{|t| - 120}{60} \right) \right], & 120 < |t| < 180, \\ 0, & |t| \geq 180 \end{cases} \quad (14)$$

For the linearly polarized light (LPL) and the single-helicity circularly polarized light (CPL), we use

$$\vec{A}_0(t) = A_0 \vec{e}_{\parallel} \cos(\Omega t) \quad (\text{LPL}), \quad (15)$$

$$\vec{A}_0(t) = \frac{A_0}{\sqrt{2}} (\cos(\Omega t), \sin(\Omega t)) \quad (\text{CPL}). \quad (16)$$

The unit vector $\vec{e}_{\parallel}$ denotes the polarization direction for the LPL, and the amplitude A_0 is given by $A_0 = E_0/\Omega$, where E_0 is the electric field amplitude and Ω is the frequency. With the solution of the EoM (11), the expectation values of the charge and spin currents are

$$\langle J_{e,\nu} \rangle(t) = \frac{1}{N_k} \sum_{\vec{k}} \text{Tr} [J_{e,\nu}(\vec{k}) \rho_{\vec{k}}(t)], \quad (17)$$

$$\langle J_{s,\mu,\nu} \rangle(t) = \frac{1}{N_k} \sum_{\vec{k}} \text{Tr} [J_{s,\mu,\nu}(\vec{k}) \rho_{\vec{k}}(t)], \quad (18)$$

where N_k is the number of momentum grid points and

$$J_{e,\nu}(\vec{k}) := -\partial_{k_{\nu}} H(\vec{k}), \quad J_{s,\mu,\nu}(\vec{k}) := \{\sigma_{\mu}, J_{e,\nu}(\vec{k})\}. \quad (19)$$

The HHGs of the charge and spin currents are obtained by the Fourier transform of these expectation values. We denote the N -th order HHG of the charge and spin current as $J_{e,\nu}^{(N)}$ and $J_{s,\mu,\nu}^{(N)}$, respectively.

Numerical result for LPL_x

We show the numerical results of HHG of the charge and spin currents for the type-I, type-II, and type-III magnets under the linearly polarized light along the x -direction (LPL_x) in Fig. 3. In this case, nonzero harmonics appear only in the longitudinal components of the charge and spin currents, because the transverse components are forbidden by the twofold rotation symmetry ($[E \parallel C_{2x}]; 0$). Since this dynamical symmetry is shared by the type-I and type-III magnets, the two phases cannot be distinguished by charge- or spin-current HHG under LPL_x.

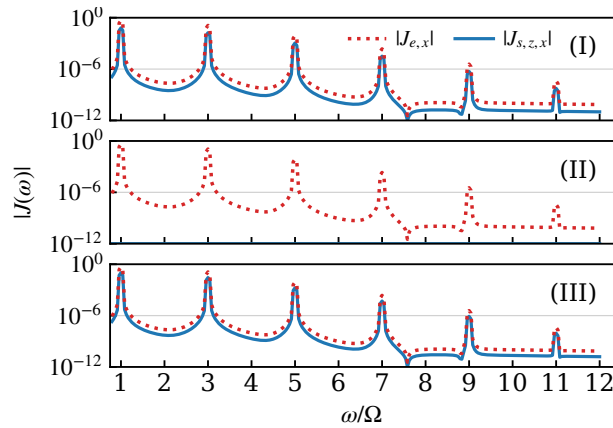


FIG. 3. HHG of the charge and spin currents for the type-I, type-II, and type-III magnets under the linearly polarized light along the x -direction (LPL_x).

TABLE II. Selection rules for HHG of the charge and spin currents under the linearly polarized light (LPL). The indices $\parallel$ and $\perp$ denote the directions parallel and perpendicular to the polarization direction, respectively.

LPL	$(J_{e,\parallel}^{(N)}, J_{e,\perp}^{(N)})$	$(J_{s,S_{\perp,\parallel}}^{(N)}, J_{s,S_{\perp,\perp}}^{(N)})$	$(J_{s,S_{\parallel,\parallel}}^{(N)}, J_{s,S_{\parallel,\perp}}^{(N)})$
$([E \parallel m_{\parallel}]; T/2)$	$(2m+1, 2m)$	$(2m+1, 2m)$	$(2m+1, 2m)$
$([C_{2x} \parallel m_{\parallel}]; T/2)$	$(2m+1, 2m)$	$(-, -)$	$(2m, 2m+1)$
$([E \parallel m_{\perp}]; 0)$	$(\forall N, -)$	$(\forall N, -)$	$(\forall N, -)$
$([C_{2x} \parallel m_{\perp}]; 0)$	$(\forall N, -)$	$(-, -)$	$(-, \forall N)$
$([E \parallel C_{2,\parallel}]; 0)$	$(\forall N, -)$	$(\forall N, -)$	$(\forall N, -)$
$([C_{2x} \parallel C_{2,\parallel}]; 0)$	$(\forall N, -)$	$(-, -)$	$(-, \forall N)$
$([E \parallel C_{2,\perp}]; T/2)$	$(2m+1, 2m)$	$(2m+1, 2m)$	$(2m+1, 2m)$
$([C_{2x} \parallel C_{2,\perp}]; T/2)$	$(2m+1, 2m)$	$(-, -)$	$(2m, 2m+1)$
$([E \parallel I]; T/2)$	$(2m+1, 2m+1)$	$(2m+1, 2m+1)$	$(2m+1, 2m+1)$
$([C_{2x} \parallel I]; T/2)$	$(2m+1, 2m+1)$	$(-, -)$	$(2m, 2m)$

TABLE III. Selection rules for HHG of the charge and spin currents under the single-helicity CPL. Here, $\parallel$ denotes the component along the single-helicity CPL propagation direction Z , while $\perp_{\pm} = X \pm iY$ denotes the circular in-plane components. The label $\pm$ denotes the eigenvalues of the rotation operator. The spin frame $S_{\parallel}$ and $S_{\perp}$ are defined as the spin direction parallel and perpendicular to the magnetic order, respectively.

CPL	$(J_{e,\parallel}^{(N)}, J_{e,\perp_{\pm}}^{(N)})$	$(J_{s,S_{\perp,\parallel}}^{(N)}, J_{s,S_{\perp,\perp_{\pm}}})$	$(J_{s,S_{\parallel,\parallel}}^{(N)}, J_{s,S_{\parallel,\perp_{\pm}}})$
$([E \parallel C_{n,\perp}]; -T/n)$	$(mn, mn \pm 1)$	$(mn, mn \pm 1)$	$(mn, mn \pm 1)$
$([C_{2x} \parallel C_{n,\perp}]; -T/n)$	$(mn, mn \pm 1)$	$(-, -)$	$(mn + n/2, mn \pm 1 + n/2)$
$([E \parallel I]; T/2)$	$(2m+1, 2m+1)$	$(2m+1, 2m+1)$	$(2m+1, 2m+1)$
$([C_{2x} \parallel I]; T/2)$	$(2m+1, 2m+1)$	$(-, -)$	$(2m, 2m)$

Selection rules

In this section, we derive the selection rules for HHG of the charge and spin currents from the dynamical symmetries. First, we determine the axes of our system under the linearly polarized light (LPL). For the spatial transformation, we use the crystalline frame denoted as $(\hat{x}_{\text{cl}}, \hat{y}_{\text{cl}}, \hat{z}_{\text{cl}})$, which is $\hat{x}_{\text{cl}} \parallel [100]$, $\hat{y}_{\text{cl}} \parallel [010]$, and $\hat{z}_{\text{cl}} \parallel [001]$. For the LPL denoted as $\vec{E}(t) = E_0 \cos(\Omega t) \vec{e}_L$, we define the laser frame $(\hat{X}, \hat{Y}, \hat{Z})$ as $\hat{X} = \vec{e}_L$, the propagation direction as $\hat{Z}$, and the perpendicular direction as $\hat{Y} = \hat{Z} \times \hat{X}$. The definition of spin frame $(\hat{S}_x, \hat{S}_y, \hat{S}_z)$ depends on whether the system symmetry is given by spin point group or magnetic point group. If spin point group is adopted, $\hat{S}_z$ is aligned with magnetic order, and $\hat{S}_x$ and $\hat{S}_y$ are defined as the perpendicular directions to the magnetic order, whereas spin frame is fixed as $(\hat{S}_x, \hat{S}_y, \hat{S}_z) = (\hat{x}_{\text{cl}}, \hat{y}_{\text{cl}}, \hat{z}_{\text{cl}})$ if magnetic point group is adopted.

Next, we define the readout components for charge and spin currents. For LPL, the charge current components are $(J_{e,X}, J_{e,Y}, J_{e,Z})$. Since the spin current transforms as the product of spin and charge-current directions, the spin current components are $(J_{s,S_x,X}, J_{s,S_x,Y}, J_{s,S_x,Z})$, $(J_{s,S_y,X}, J_{s,S_y,Y}, J_{s,S_y,Z})$, and $(J_{s,S_z,X}, J_{s,S_z,Y}, J_{s,S_z,Z})$.

Selection rules for charge current and spin current with spin point group

We consider the dynamical symmetries with the spatial transformations $(m_{\parallel}; T/2)$, $(m_{\perp}; 0)$, $(C_{2,\parallel}; 0)$, and $(C_{2,\perp}; T/2)$, where the indices $\parallel$ and $\perp$ denote the directions of mirror planes or rotation axes parallel and perpendicular to the polarization direction, respectively. From these dynamical symmetries, we can derive the selection rules for HHG of the charge currents, which are summarized in Table II.

In the case of spin point group with collinear magnet ordered along the z -axis, the spin frame is simplified as $(\hat{S}_{\perp}, \hat{S}_{\parallel})$, where $\hat{S}_{\parallel} = \hat{S}_z$ is the direction of magnetic order and $\hat{S}_{\perp}$ is the direction perpendicular to the magnetic order. From this definition, by the spin space rotation, these spin directions are transformed as $(\hat{S}_{\perp}, \hat{S}_{\parallel}) \rightarrow (\hat{S}_{\perp}, -\hat{S}_{\parallel})$ or $(\hat{S}_{\perp}, \hat{S}_{\parallel}) \rightarrow (-\hat{S}_{\perp}, -\hat{S}_{\parallel})$, depending on the gauge choice of the spin space rotation.

Table III summarizes the selection rules for HHG of the charge and spin currents under the single-helicity circularly polarized light (CPL). In this table, we define the basis $\perp_{\pm} = \hat{X} \pm i\hat{Y}$, which are eigenvectors of the rotation operator. A key feature is that the selection rules for spin currents imposed by rotation are identical to those for charge currents, which is different from the case of LPL. The second row, which contains the shift $n/2$ from the spin flip, assumes even n .

TABLE IV. Selection rules for HHG of spin currents under magnetic point group symmetries for LPL polarized along [100] direction. The indices $\parallel$ and $\perp$ denote the directions parallel and perpendicular to the polarization direction, respectively.

MPG	$(J_{s,S_X,\parallel}^{(N)}, J_{s,S_X,\perp}^{(N)})$	$(J_{s,S_Y,\parallel}^{(N)}, J_{s,S_Y,\perp}^{(N)})$	$(J_{s,S_Z,\parallel}^{(N)}, J_{s,S_Z,\perp}^{(N)})$
$(m_{\parallel}; T/2)$	$(2m+1, 2m)$	$(2m, 2m+1)$	$(2m, 2m+1)$
$(m_{\perp}; 0)$	$(-, {}^{\vee}N)$	$({}^{\vee}N, -)$	$(-, {}^{\vee}N)$
$(C_{2,\parallel}; 0)$	$({}^{\vee}N, -)$	$(-, {}^{\vee}N)$	$(-, {}^{\vee}N)$
$(C_{2,\perp}; T/2)$	$(2m, 2m+1)$	$(2m, 2m+1)$	$(2m+1, 2m)$
$(I; T/2)$	$(2m+1, 2m+1)$	$(2m+1, 2m+1)$	$(2m+1, 2m+1)$

TABLE V. Selection rules for HHG of spin currents under magnetic point group symmetries for LPL polarized along [110] direction. The indices $\parallel$ and $\perp$ denote the directions parallel and perpendicular to the polarization direction, respectively.

MPG	$(J_{s,S_{\parallel+},\parallel}^{(N)}, J_{s,S_{\parallel+},\perp}^{(N)})$	$(J_{s,S_{\parallel-},\parallel}^{(N)}, J_{s,S_{\parallel-},\perp}^{(N)})$	$(J_{s,S_Z,\parallel}^{(N)}, J_{s,S_Z,\perp}^{(N)})$
$(m_{\parallel}; T/2)$	$(2m+1, 2m)$	$(2m, 2m+1)$	$(2m+1, 2m)$
$(m_{\perp}; 0)$	$(-, {}^{\vee}N)$	$({}^{\vee}N, -)$	$(-, {}^{\vee}N)$
$(C_{2,\parallel}; 0)$	$({}^{\vee}N, -)$	$(-, {}^{\vee}N)$	$(-, {}^{\vee}N)$
$(C_{2,\perp}; T/2)$	$(2m, 2m+1)$	$(2m, 2m+1)$	$(2m+1, 2m)$
$(I; T/2)$	$(2m+1, 2m+1)$	$(2m+1, 2m+1)$	$(2m+1, 2m+1)$

Selection rules for HHG with magnetic point group

For the case of magnetic point group, the spin frame is fixed as $(\hat{S}_x, \hat{S}_y, \hat{S}_z) = (\hat{x}_{cl}, \hat{y}_{cl}, \hat{z}_{cl})$. In this case, the relative direction between the laser frame and crystalline frame is important, and the selection rules depend on the relative direction. In Tables IV and V, we show the selection rules for HHG of the charge and spin currents under the LPL from magnetic point group symmetries. From Table IV, one can see that some selection rules obtained from the SPG are reproduced by the magnetic point group.

For single-helicity CPL, we derive the selection rules for spin current HHG. Since spin is not affected by inversion, the selection rules are the same as those for the result with SPG. Thus, the remaining symmetry preserving the CPL helicity and leading to non-trivial results is the n -fold rotation C_n . When the rotation axis is parallel to the propagation direction of the CPL, we have the dynamical symmetry $(C_n; -T/n)$. In this situation, we use the in-plane eigenbasis $S_{\alpha} = S_X + i\alpha S_Y$ and $J_{\beta} = J_X + i\beta J_Y$ with $\alpha, \beta = \pm 1$, and denote the out-of-plane components by S_Z and J_Z . The relevant spin-current components are $J_{s,S_{\alpha},\beta}^{(N)}$, $J_{s,S_Z,\beta}^{(N)}$, $J_{s,S_{\alpha},Z}^{(N)}$, and $J_{s,S_Z,Z}^{(N)}$. For these spin current bases, one obtains

$$\left(1 - e^{-2i\pi N/n} e^{2i\pi(\alpha+\beta)/n}\right) J_{s,S_{\alpha},\beta}^{(N)} = 0, \quad \left(1 - e^{-2i\pi N/n} e^{2i\pi\beta/n}\right) J_{s,S_Z,\beta}^{(N)} = 0, \quad (20)$$

$$\left(1 - e^{-2i\pi N/n} e^{2i\pi\alpha/n}\right) J_{s,S_{\alpha},Z}^{(N)} = 0, \quad \left(1 - e^{-2i\pi N/n}\right) J_{s,S_Z,Z}^{(N)} = 0, \quad (21)$$

Therefore, the allowed harmonics are $N = nm + \alpha + \beta$, $N = nm + \beta$, $N = nm + \alpha$, and $N = nm$ for $J_{s,S_{\alpha},\beta}^{(N)}$, $J_{s,S_Z,\beta}^{(N)}$, $J_{s,S_{\alpha},Z}^{(N)}$, and $J_{s,S_Z,Z}^{(N)}$, respectively. The key point is that no magnetic-point-group operation reproduces the SPG spin-current selection rule under a fixed single-helicity CPL. To reproduce the selection rules generated by $[C_2 \parallel C_n]$, one would need to consider the antiunitary magnetic point group operation $C_n\mathcal{T}$. However, this operation flips the helicity of the CPL and therefore cannot be a dynamical symmetry of the same single-helicity CPL. Thus, fixed-helicity CPL distinguishes an altermagnet with nonrelativistic spin splitting from a system with relativistic spin splitting.

Parameter settings

The parameters for the numerical calculations are set as follows. For the band plot in Fig. 2, we use $a = 1.0$, hopping parameters $t = 1.0$ and $t' = 0.3$, $\mu = 0.5$, and $\Delta = M = 0.3$. For the HHG simulations, we use a uniform 501×501 momentum mesh over the full Brillouin zone $[-\pi/a, \pi/a]^2$. The model parameters are $a = 1.0$, hopping parameters $t = 1.0$ and $t' = 0.3$, $\Delta = 0.3$ for the type-I magnet, and $M = 0.3$ for the type-II and type-III magnets. The equilibrium density matrix is evaluated with inverse temperature $\beta = 20$ and chemical potential $\mu = 1.7$. The relaxation time is $T_2 = 120$. The chemical potential used for the band visualization is chosen for clarity, while the HHG simulations use $\mu = 1.7$; the selection rules are symmetry constraints and are independent of this choice as long as the equilibrium state preserves the assumed symmetries.

The time evolution is performed from $t_{\text{time}} = -200$ to 200 with time step $\Delta t = 0.002$, giving 200001 time points. We use a fourth-order Runge–Kutta integrator for Eq. (11). The external field parameters are $E_0 = 1.5$ and $\Omega = 0.7$, so that $A_0 = E_0/\Omega$. For the LPL_[110] production run, the ellipticity is $\epsilon = 0$ and the polarization angle is $\theta = \pi/4$. The pulse envelope is the flat-top envelope defined in the time-evolution section, with ramp time 60 and plateau duration 240.

The current expectation values are evaluated as equal-weight averages over the Brillouin zone. The charge current operator is $J_{e,\nu} = -\partial_{k_\nu} H$, and the spin current operator used in the production calculation is the anticommutator $J_{s,\mu,\nu} = \{\sigma_\mu, J_{e,\nu}\}$ without an additional factor of $1/2$. The harmonic spectra are computed from the time window $t_{\text{time}} \in [-107.712, 107.712]$ after subtracting the mean value and applying a Hann window. This Fourier window contains 107713 time points, corresponding to approximately 24 driving periods and angular-frequency resolution 0.02916632768.